

Pumping Lemma for Regular Language

- For showing certain languages not to be regular

Theorem:

→ Let L be a regular language.

→ Then there exists a constant n such that for every string w in L ($|w| \geq n$).

→ we can break w into 3 strings, $w = xyz$ such that

1. $y \neq \epsilon$ (or) $|y| > 0$

2. $|xy| \leq n$

3. For all $k \geq 0$, then string xy^kz is also in L .

1) Prove that $L = \{0^n 1^n \mid n \geq 1\}$, is not Regular

* Assume L is Regular

- Let n be a constant

- Let $w = 0^n 1^n$, $|w| \geq n$

- Split $w = xyz$ such that,

(1) $|y| \neq \epsilon$

2) $|xy| \leq n$

3) for all $k \geq 0$ $xy^kz \in L$.

$w = 0^n 1^n$

$n = 2$ $w = 0011$ $N = 4$

$x = 0$ $y = 01$ $z = 11$

(i) $|y| = \epsilon$ $|y| = 2$ (T) $2 \leq 2$ (T)

(ii) $|xy| \leq n$ $|xy| = 2$ ~~$2 \leq 4$ (T)~~

(iii) $xy^kz \in L$ $\frac{k=2}{xy^2z} \Rightarrow 00011 \notin L$ (F)

$xy^2z \Rightarrow 00011 \notin L$ (F)

Hence, Language is not Regular

Ex 2 $L = \{a^i b^j \mid i \leq j\}$ is not regular.
 $\{aabb, aabbb, abb\}$

* Assume that L is Regular.

* Let n be number of states of DFA

* Let $w = a^n b^{n+1}$, $|w| > n$

* Split $w = xyz$ such that

1) $y \neq \epsilon$

2) $|xy| \leq n$

3) For all $k \geq 0$, $xy^k z \in L$

Let $n \geq 2$ $w = aabbb$.

$$xy = aa$$

$$x = a \quad y = a \quad z = bbb$$

(i) $y \neq \epsilon$ (✓)

(ii) $|xy| \leq 2 \quad 2 \leq 2$ (✓)

(iii) $xy^k z \in L \quad k \geq 2$

$$xy^2 z \Rightarrow aabbbb \in L$$

$$k = 3$$

$$xy^3 z = aaabbbb \notin L.$$

Contradiction!

Hence, L is not Regular